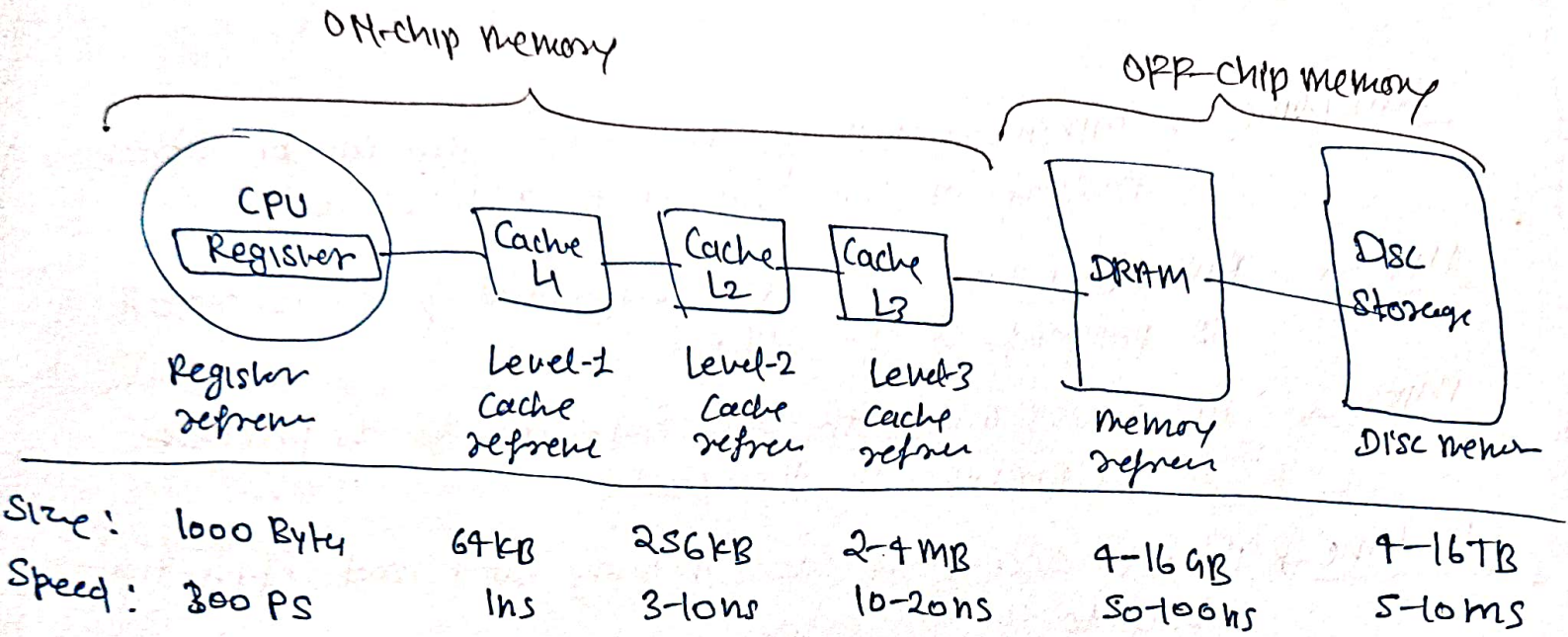
Cache memory

→ Create an illusion of very large and fast memory.
→ multiple level of memory with different speeds and sizes



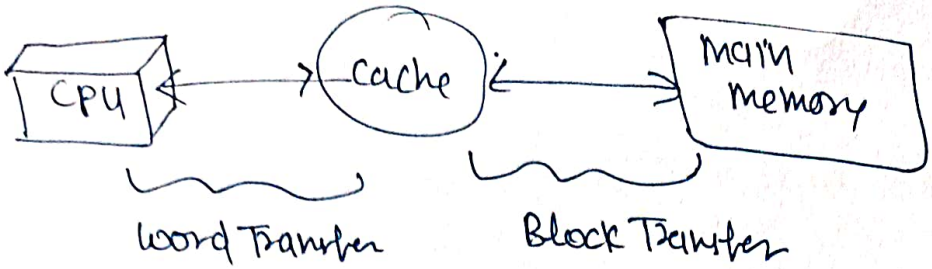
Size:	1000 Bytes	64KB	256KB	2-4MB	4-16GB	4-16TB
Speed:	300 PS	1ns	3-10ns	10-20ns	50-100ns	5-10ms

Cache is small, Fast buffer between processor and memory.
→ Old value will be removed from cache to make space for new values.

Principal of locality → Program Access a relatively small portion of their address space at any instant of time.

Temporal locality → If an item is referenced, it will tend to be referenced again soon

Spatial locality → If an item is referenced, items whose addresses are closed by will tend to be refereneed soon.



Block/line → Minimum unit of information that can be either present or not present in a cache level.

Hit → An access where the data requested by the processor is present in the cache.

Miss → An access where the data requested by the processor is not present in the cache

Hit time → Time to access the cache memory block and return the data to the processor

Hit Rate / miss Rate → Fraction of memory access found (not found) in the cache.

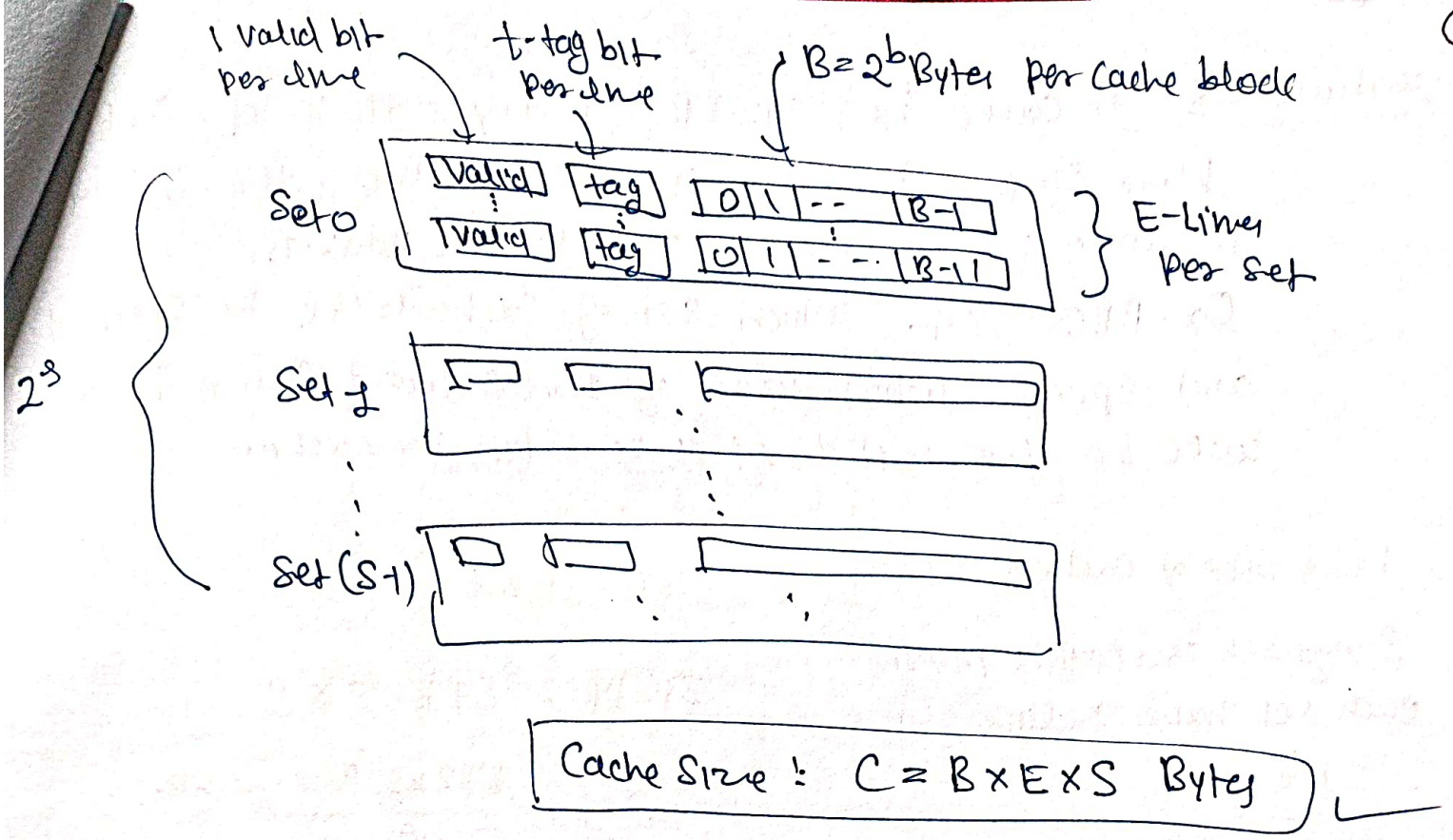
Miss penality → Time to replace a block in the cache with the corresponding block from the next level. (Additional cycle require to take data in cache)

General Organization of cache

→ Cache is an array of sets

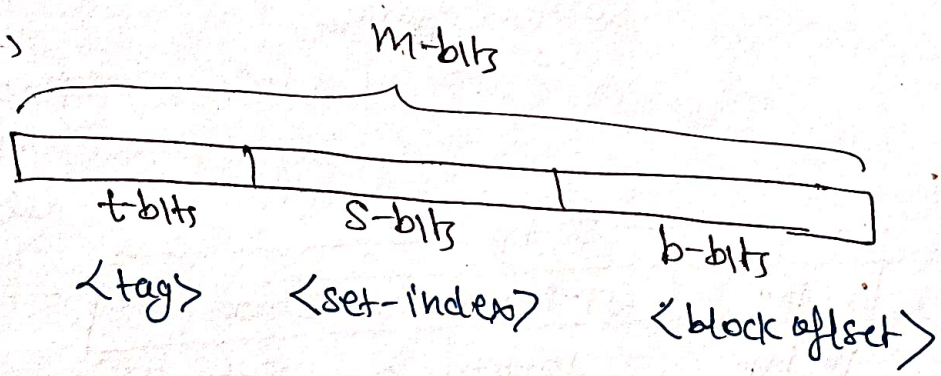
→ Each set contains one or more lines

→ Each line holds a block of data



Cache Size : $C = B \times E \times S$ Bytes

Addressing Cache →



- Locate the set based on <set-index>
- Locate the line in the set based on <tag>
- check that the line is valid
- Locate the data in the line based on <block offset>

Question → A cache has 512 KB capacity, 4B word, 64B block size and 8-way set associative. the system is using 32-bit address. Given the Address 0x·ABC89984. which set of cache will be selected and specify which word of the selected cache block will be forwarded if it is a hit in cache?

Total Size of Cache = 512 KB,

$C = B \times E \times S$

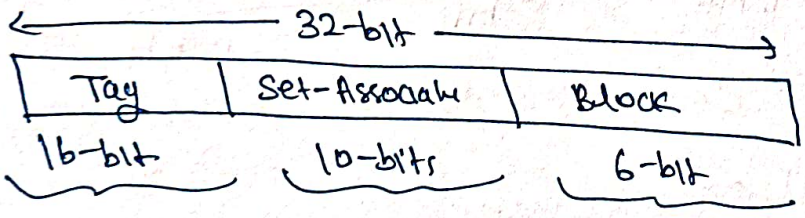
8-way set associative means each set have 8-lines

$E = 8$

$512\,KB = 64B \times 8 \times S$

$S = \frac{512 \times 2^{10}}{2^9} = 2^{10}$

So total Number of set are = 1024, require number of bit to represent all sets is = 10 bits.



$64B = 2^6 B$

$6 = 4 + 2$

↓

word Number

$1W = 4B$

$W = \frac{64B}{4B} = 16$ words

Address →

0x ABC89984

↑ tag bit (16-bit)

→ 1001 1001 1000 0100

10-bit 4-bit 2-bit

↓

set Number = 614, word = 1

Four Cache memory design choise :

- Where can be block be placed in the cache?
  → Block placement
- How is a block found If it is in the upper level?
  → Block Identification
- Which block should be replaced on a miss?
  → Block Replacement
- What happen on a write?
  → write Strategy

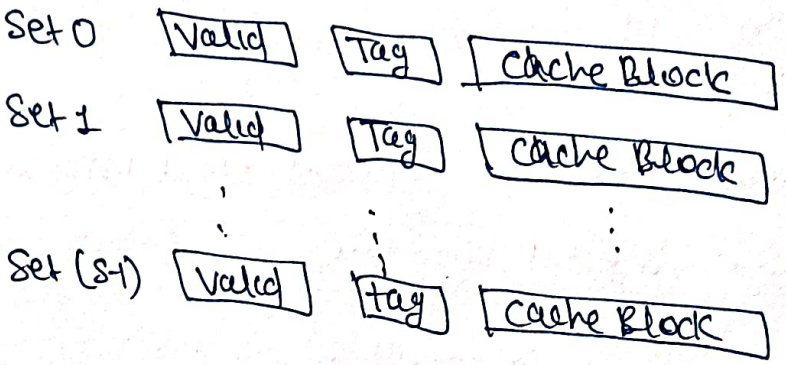
Cache Mapping → (Block placement)

1. Direct mapping
   - Block can be placed in only one location
   - (Block Number) mod (Number of blocks in cache)
   - $\frac{12}{8} = 4$
2. Set-Associative →
   - Block can be placed in one among a list of location
   - (Block Number) mod (Number of sets)
3. Fully Associative → Block can be placed anywhere.

Direct Mapping Cache →

- Simplest kind of cache, easy to build
- only 1 tag compare required per access.
- characterized by exactly one line per set

Cache size (C) = B × S Bytes

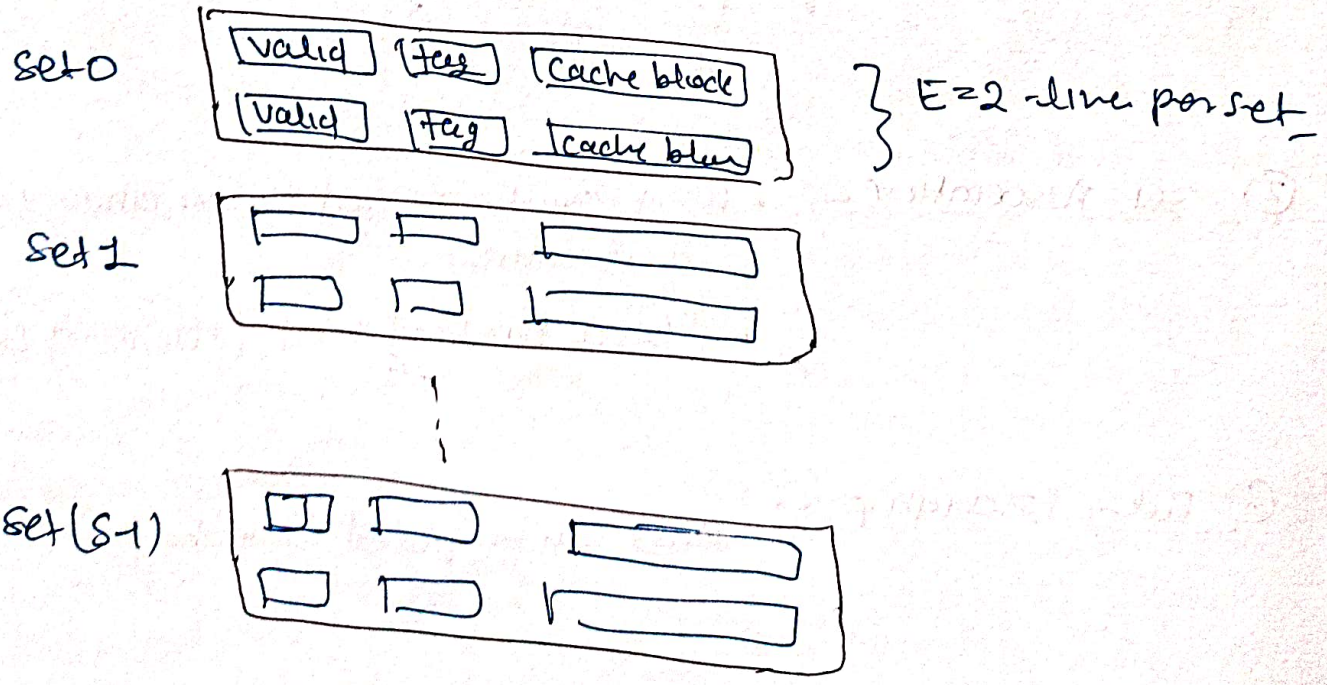


→ Block matching: Find a valid block in the selected set with a matching tag.

→ Word selection: Then extract the word.

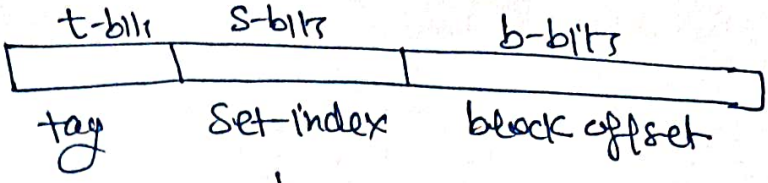
Set-Associative Cache →

characterized by more than one line per set.



2-way set associative

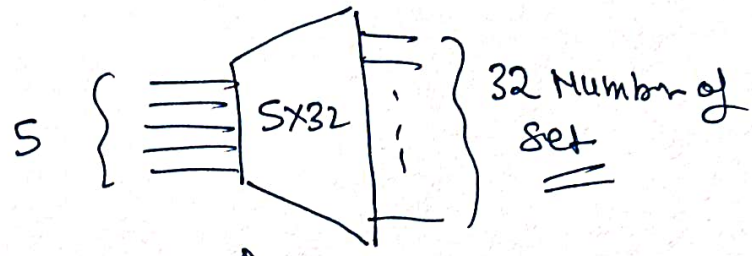
Cache Indexing –



↓

Let (00001) – s-bit

- Decoder are used for Indexing
- Indexing time depends upon decoder size (S:2S)
- Smaller Number of Sets, less Indexing time.



Decoder